Smart Campus Sustainability

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certfied that 21CSC203P-Advance programming practice report titled "**Smart Campus Sustainability and Carbon footprint monitoring System"** is the bonafide work of **Yajat M rawal-RA2511030040010,S SURENDAR-RA2511030040005,SURYA SWAMINATHAN-RA2511030040008,SHIVRAJ SINGH-RA2611030040021,**who carried the project under my supervision.Certified further,that to the best of my knowledge the work reported here in does not form an any other project report dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Universities and colleges behave like small cities: they consume electricity, water and fuel at scale, generate waste and move thousands of people every day, yet most institutions measure their environmental impact only once a year from utility bills. Annual figures are too late and too coarse to change behaviour or operations, and faults such as leaking tanks or air-conditioning left running overnight go unnoticed for weeks.

This project designs and implements **EcoSphere AI**, a Smart Campus Sustainability and Carbon Footprint Monitoring System. Readings from smart meters and IoT sensors are ingested through an authenticated gateway API, validated, stored and converted into greenhouse-gas emissions by a carbon engine that follows the GHG Protocol (Scope 1, 2 and 3) using version-controlled emission factors such as the CEA grid factor. An analytics layer learns each building's normal load profile and raises alerts for after-hours consumption, demand spikes, night-time water leaks, offline or stuck meters and poor indoor ventilation. A regression model forecasts daily electricity demand from the academic calendar and temperature, a benchmarking module ranks buildings by Energy Performance Index, and a scenario model estimates savings from LED retrofits, rooftop solar, set-point changes, occupancy controls and electric buses before money is spent.

The system is built entirely with free and open-source software: Python, Django REST Framework, SQLite, NumPy and ReportLab on the server, and React, TypeScript, Recharts and Leaflet in the browser. Role-based dashboards serve management, the facility team and the student community, and a reporting module produces GHG inventory PDFs and CSV exports.

Evaluated on a 120-day simulated dataset for a ten-building campus of 5,850 people, the platform reported an annualised footprint of about 7,975 tCO_2e (1.53 t per student), detected all six injected faults with no false alarms, and forecast daily demand with a mean absolute percentage error of 2.36%. A combined package of measures was estimated to avoid about 797 tCO_2e per year with a payback of 3.1 years. The results show that continuous monitoring can turn sustainability from an annual report into a daily operating practice.

Keywords: smart campus, carbon footprint, GHG Protocol, IoT, anomaly detection, energy forecasting, Django, React, sustainability dashboard.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BEE	Bureau of Energy Efficiency
CEA	Central Electricity Authority
CO_2e	Carbon dioxide equivalent
EPI	Energy Performance Index
GHG	Greenhouse Gas
HVAC	Heating, Ventilation and Air Conditioning
IoT	Internet of Things
JWT	JSON Web Token
KPI	Key Performance Indicator
MAD	Median Absolute Deviation
MAPE	Mean Absolute Percentage Error
REST	Representational State Transfer
RBAC	Role-Based Access Control
STP	Sewage Treatment Plant
tCO_2e	Tonnes of carbon dioxide equivalent

CHAPTER 1
INTRODUCTION

1.1 Background

Higher-education campuses are significant contributors to local energy demand and emissions. A typical campus includes academic blocks, hostels, laboratories, libraries, canteens, sports facilities, staff housing and a vehicle fleet. In warm and humid regions such as Chennai, cooling loads make grid electricity the dominant source of emissions, and water is pumped, stored and treated on site at a further energy cost.

At the same time, campuses are well placed to lead. They have a captive population, control over their own infrastructure, research capability and an educational mission. A monitored, data-driven campus doubles as a living laboratory: the same data that helps the estate office find a leaking valve can be used by students for coursework in data science, energy systems and environmental studies.

The guiding principle of this project is simple: *what is measured gets managed*. Continuous monitoring turns sustainability from an annual compliance exercise into a daily operating practice.

1.2 Problem Statement

Current campus sustainability practice commonly suffers from the following gaps:

- Energy and water are recorded at the main meter only, so individual buildings and departments are invisible and cannot be held accountable.
- Carbon accounting is manual, periodic and spreadsheet-based, which makes it slow and error-prone.
- Faults such as leaks, idle air-conditioning and lights left on go unnoticed for days or weeks.
- Students and staff receive no feedback on the impact of their behaviour.
- Reports are hard to compare year on year because methods, factors and boundaries change.

1.3 Objectives

1. Monitor electricity, solar generation, water, indoor environment, waste and transport data in near real time.
2. Calculate campus emissions automatically using recognised, version-controlled emission factors aligned with the GHG Protocol.
3. Detect anomalies and wastage and route alerts to the responsible team as trackable work orders.
4. Provide role-specific dashboards for management, facility staff and the campus community.
5. Support target setting, demand forecasting, benchmarking and what-if scenario analysis.
6. Build a scalable, low-cost architecture using only free and open-source software that can start small and expand building by building.

1.4 Scope of the Project

The system covers measurement, data transmission, storage, analytics, visualisation, alerting and reporting. It addresses Scope 1 (generators, fleet, LPG, refrigerant leakage), Scope 2 (purchased electricity, location- and market-based) and a practical subset of Scope 3 (water supply and treatment, waste disposal and student and staff commuting). It does not replace existing building management systems; it integrates with them through a documented ingestion API. Because physical meters were not available during development, a physics-based simulator generates realistic sensor data so that

every layer of the system can be exercised end to end.

1.5 Organisation of the Report

Chapter 2 reviews the standards and existing approaches that informed the design. Chapter 3 lists the functional and non-functional requirements. Chapter 4 presents the system architecture, data flow and database design, and Chapter 5 explains the carbon accounting method and the analytical algorithms. Chapter 6 describes the implementation, Chapter 7 discusses the results, Chapter 8 covers testing, and Chapters 9 and 10 present limitations, future enhancements and conclusions.

CHAPTER 2
LITERATURE SURVEY AND STANDARDS

2.1 Carbon Accounting Standards

The **GHG Protocol Corporate Accounting and Reporting Standard** is the most widely used framework for organisational inventories. It classifies emissions into three scopes and requires the organisation to define its boundary (operational or financial control) and a base year against which progress is measured. Its Scope 2 guidance requires dual reporting: a *location-based* figure using the average grid factor and a *market-based* figure that reflects contractual purchases of renewable power.

ISO 14064-1 specifies principles for quantifying and reporting GHG emissions and removals, while **ISO 50001** defines energy-management systems built around Energy Performance Indicators (EnPIs) and baselines. For India, the **Central Electricity Authority (CEA) CO_2 Baseline Database** publishes the grid emission factor, recently in the region of 0.7-0.8 kg CO_2/kWh; because the value is revised periodically, a monitoring system must store factors as dated, versioned parameters.

2.2 Campus Sustainability Rating Systems

Green-building rating systems such as **GRIHA** and **IGBC** assess energy performance (kWh/m^2/year), water efficiency, waste management and renewable energy. **AASHE STARS**, used by universities worldwide, adds operations, engagement and curriculum credits, and NIRF-style disclosures in India increasingly ask for sustainability data. All of them reward measured, verifiable data over estimates.

The UN Sustainable Development Goals most relevant to this work are SDG 7 (affordable and clean energy), SDG 11 (sustainable cities), SDG 12 (responsible consumption) and SDG 13 (climate action).

2.3 Existing Approaches

Table 2.1: Comparison of existing approaches with the proposed system

Approach	Strengths	Limitations
Annual utility-bill audit	Cheap, credible totals	No building detail; results arrive months late
Building Management System (BMS)	Real-time control of HVAC	Vendor-specific; no carbon accounting or community view; expensive licences
Commercial energy dashboards	Polished visualisation	Subscription cost; electricity only; data held by vendor
Spreadsheet carbon calculators	Flexible, familiar	Manual, error-prone, no alerts, poor versioning
Proposed system	Integrated multi-resource monitoring, GHG engine, alerts, forecasting and engagement; free software	Requires metering investment for real deployment

2.4 Analytical Techniques

Research on building energy analytics shows that simple, explainable models perform well for campus loads, which are driven mainly by occupancy schedules and weather. Robust statistics such as the median and median absolute deviation (MAD) are widely recommended for anomaly detection because, unlike the mean and standard deviation, they are not distorted by the very anomalies being sought. Night-time minimum flow analysis is the standard method used by water utilities to detect leaks, and multiple linear regression with calendar and temperature terms is a common baseline for short-term demand forecasting. These findings guided the choice of transparent algorithms that can be explained, audited and run on modest hardware.

CHAPTER 3
REQUIREMENTS ANALYSIS

3.1 Stakeholders and Roles

Table 3.1: User roles and access rights

Role	Typical users	Capabilities
Administrator	Sustainability officer, management	All dashboards; campus settings and boundary; emission factors; targets; buildings, meters and gateways; user roles
Facility team	Estate office, electricians, plumbers	Operations console; acknowledge and resolve alerts; run anomaly scans; log fuel, refrigerant and waste
Community	Students and staff	Read-only dashboards; hostel and department leaderboards; private commute footprint calculator

3.2 Functional Requirements

Table 3.2: Functional requirements

ID	Requirement
FR1	Ingest meter readings from authenticated gateways; reject malformed data and flag out-of-range values.
FR2	Convert activity data to kg CO_2e by scope, source, building and period using the factor valid on each day.
FR3	Show campus KPIs: total and per-student emissions, kWh/m^2, renewable share, water per person, waste diversion and progress to target.
FR4	Detect after-hours load, spikes, leaks, stuck or offline meters and high CO_2, and create de-duplicated alerts with estimated cost.
FR5	Allow alerts to be acknowledged, annotated, assigned and resolved (work-order tracking).
FR6	Forecast daily electricity demand, emissions and cost with a prediction interval.
FR7	Benchmark buildings and model the savings, cost and payback of interventions.
FR8	Record waste weighings, fuel and refrigerant use, trees and commute surveys.
FR9	Generate a GHG inventory PDF and CSV exports for any date range.
FR10	Register users, authenticate with tokens and enforce role-based permissions.

3.3 Non-Functional Requirements

- **Cost:** every component must be free and open source; no paid APIs or cloud subscriptions.
- **Security:** hashed passwords, JWT access tokens with refresh rotation, per-gateway device keys, role-based authorisation and rate limiting on public endpoints.
- **Data integrity:** suspicious values are flagged, never silently corrected or deleted; factor history is append-only so past results remain reproducible.
- **Privacy:** commute answers are visible only to the person who submitted them; only aggregated totals are reported.
- **Performance:** dashboard queries should return in under one second for a year of hourly data on a laptop.
- **Usability and portability:** responsive interface for desktop and mobile; runs on Windows, Linux or macOS with a one-step setup.

3.4 Hardware and Software Requirements

Table 3.3: Technology stack

Layer	Choice (all free)	Purpose
Backend language	Python 3.12	Services, analytics and management commands
Web framework	Django 5.2, Django REST Framework 3.17	REST API, ORM, admin site
Authentication	djangorestframework-simplejwt	JSON Web Tokens with refresh rotation
Database	SQLite (PostgreSQL/TimescaleDB ready)	Readings, metadata, factors and logs
Numerics and reports	NumPy, ReportLab	Regression forecasting; PDF generation
Frontend	React 19, TypeScript, Vite	Single-page application
UI libraries	TanStack Query, Zustand, Recharts, Leaflet + OpenStreetMap	Data fetching, state, charts, campus map
Hardware (deployment)	ESP32 boards, Modbus meters, ultrasonic flow meters	Low-cost sensing
Development machine	Any PC with 4 GB RAM	Runs all services locally

CHAPTER 4
SYSTEM DESIGN

4.1 Layered Architecture

The system follows the five-layer architecture proposed in the project report. Each layer has a single responsibility and communicates with its neighbours through well-defined interfaces, so that, for example, a simulated meter can be replaced by a real Modbus meter without any change above the connectivity layer.

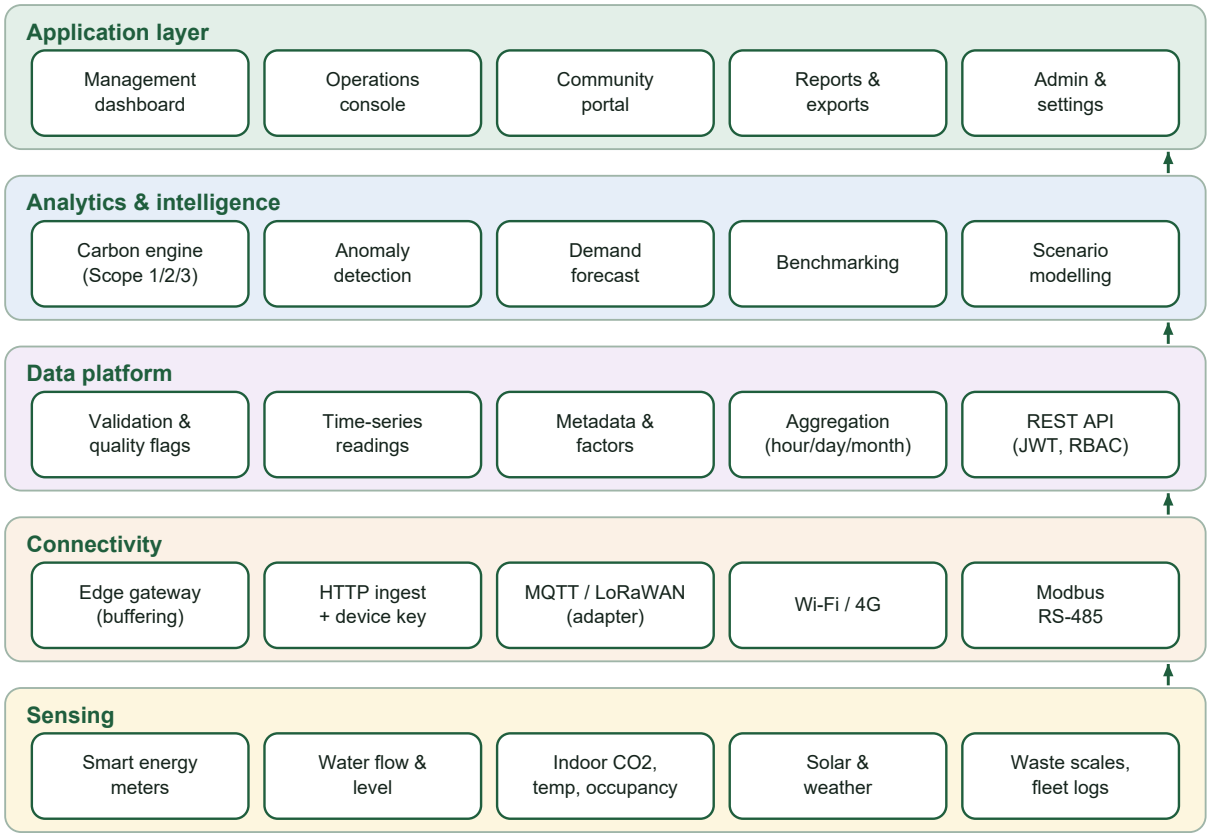


Figure 4.1: Five-layer architecture of the system

- **Sensing:** energy meters at each building incomer, rooftop solar meters, water meters per building and per source, indoor environment sensors, a weather station, waste scales and fleet logs.
- **Connectivity:** edge gateways buffer readings during outages and forward them over HTTP with a secret key; MQTT or LoRaWAN can be bridged through the same adapter.
- **Data platform:** validation, quality flags, time-series storage and aggregation to hourly, daily and monthly values; metadata such as buildings, tariffs and factors in relational tables.
- **Analytics:** carbon engine, anomaly detection, forecasting, benchmarking and scenario modelling.
- **Application:** management dashboard, operations console, community portal and reporting.

4.2 Data Flow

Figure 4.2 shows how a reading travels through the system. Devices push readings to the gateway, which authenticates with the ingestion endpoint. Valid readings are stored; out-of-range readings are stored with a *suspect* flag and excluded from totals. Analytics services read the store on demand when a dashboard calls the REST API, and a periodic scan writes alerts that appear in the operations console.

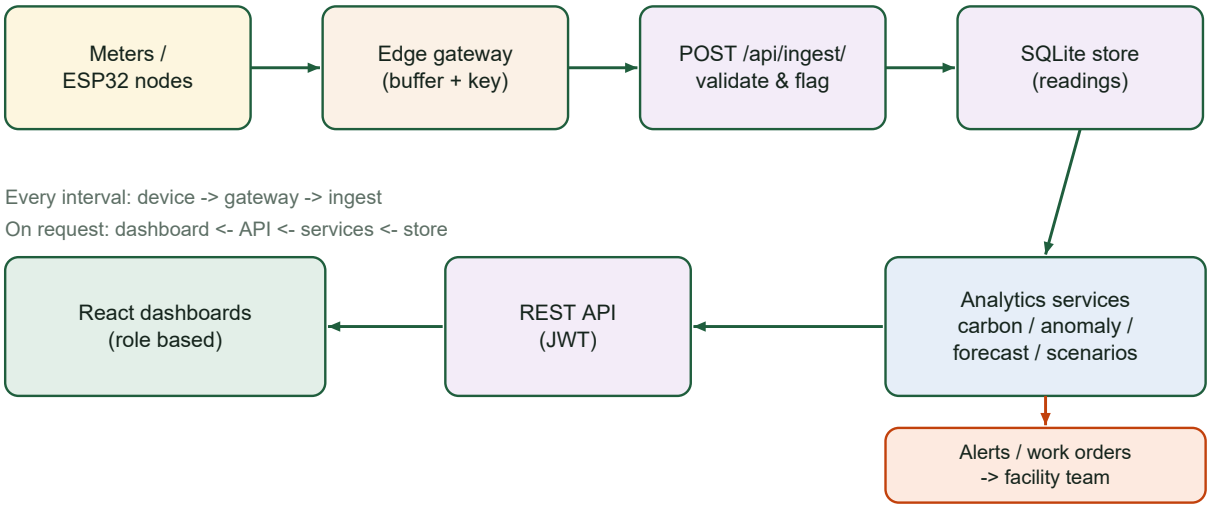


Figure 4.2: Data flow from device to dashboard

4.3 Database Design

The relational schema separates the physical campus (buildings, gateways, meters), measurements (readings, indoor environment, weather), logged activity data (fuel, waste, commuting, trees), configuration (emission factors, targets, calendar, campus settings) and operations (alerts, users). Readings are indexed on (meter, timestamp) so that hourly and daily aggregation stays fast.

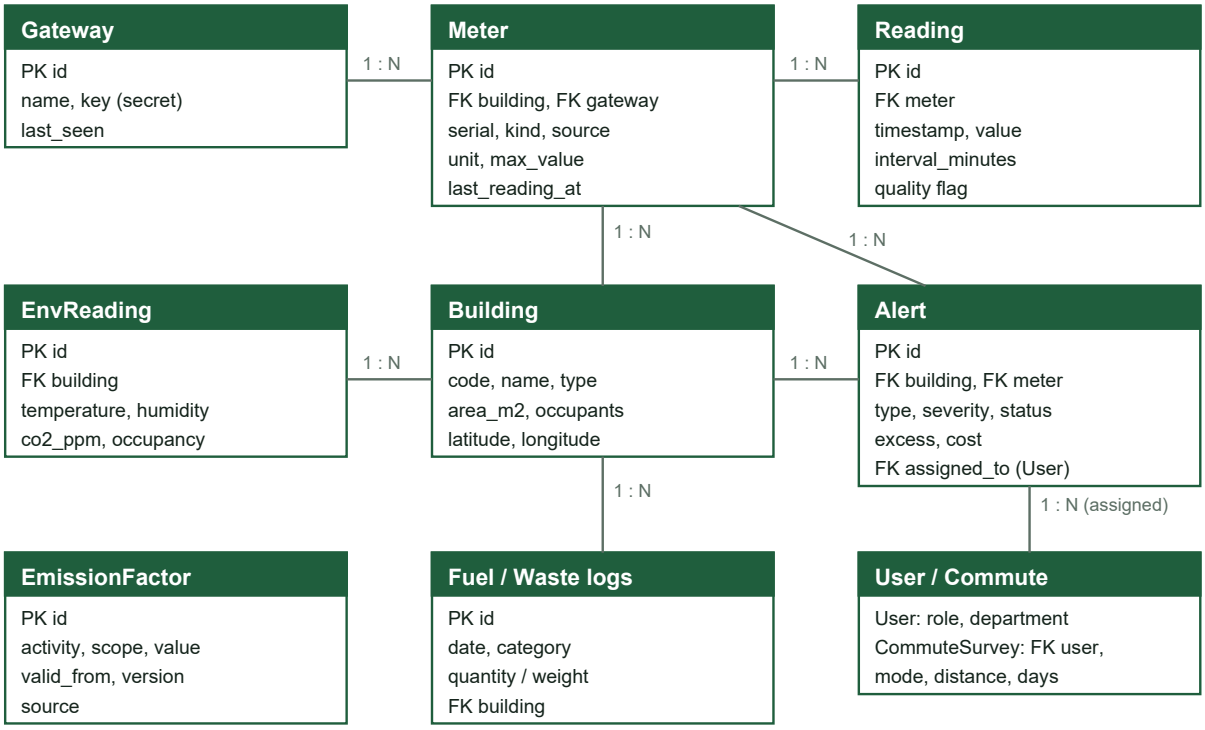


Figure 4.3: Simplified entity-relationship diagram

4.4 API Design

All functionality is exposed through a versionable REST API under `/api/`. The main endpoints are listed below; further CRUD endpoints exist for buildings, meters, gateways, factors, logs, trees, targets and users.

Table 4.1: Key REST API endpoints

Endpoint	Method	Description
/api/auth/login/, /refresh/, /register/	POST	Obtain and rotate JWT tokens; self-registration
/api/ingest/	POST	Gateway ingestion (X-Gateway-Key header)
/api/analytics/overview/	GET	Campus KPIs, scope split, monthly trend, target progress
/api/analytics/live/	GET	Current demand, solar, water flow and indoor status per building
/api/analytics/forecast/	GET	Demand forecast with 95% interval and model metrics
/api/analytics/scenarios/	POST	Savings, cost and payback for chosen interventions
/api/alerts/{id}/	PATCH	Acknowledge, assign, annotate or resolve an alert
/api/reports/ghg.pdf	GET	GHG inventory report for a date range

4.5 Security Design

Passwords are stored with Django's PBKDF2 hasher. Users authenticate once and receive a short-lived access token (30 minutes) and a refresh token (7 days) that is rotated on use. Every API view declares a permission class: read-only for all signed-in users, write access restricted to administrators or facility staff as appropriate. Devices never use user accounts; each gateway has its own random 160-bit key that can be revoked independently. Anonymous endpoints are rate-limited, and administrators cannot remove their own privileges by mistake.

CHAPTER 5
METHODOLOGY AND ALGORITHMS

5.1 Carbon Accounting Method

Every emission line is calculated with the basic relationship of the GHG Protocol:

$$E_d\,(kg\ CO_2e) = A_d \times EF_d$$

where A_d is the activity data on day d (kWh, litres, kg, kL or passenger-km) and EF_d is the emission factor whose *valid_from* date is the latest on or before d. Applying factors day by day means that publishing a new CEA value changes only the days after it takes effect, and every historical report can be reproduced exactly. Market-based Scope 2 is computed as the location-based value multiplied by (1 - certified green power share).

Commuting is estimated from anonymous surveys. For respondents i using mode m, the daily passenger-km scaled to the whole population P from n responses is:

$$km_m = (P / n) \times (44 / 52) \times \Sigma_{i\ in\ m}\ 2 \cdot dist_i \cdot days_i / 7$$

where 44/52 spreads the academic year over the calendar. Tree sequestration is calculated separately and never subtracted from emissions, following the conservative approach recommended in the

project report.

Table 5.1: Default emission factors (to be confirmed before disclosure)

Activity	Scope	Factor	Source (indicative)
Grid electricity	2	0.716 kg/kWh (v2, from Apr 2025)	CEA CO_2 Baseline Database
Diesel (generators, buses)	1	2.68 kg/L	IPCC 2006 / DEFRA
LPG (canteen, labs)	1	2.98 kg/kg	IPCC 2006
Refrigerant R-410A / R-32	1	2088 / 675 kg/kg	IPCC AR4 GWP-100
Water supply and treatment	3	0.344 kg/kL	DEFRA (indicative)
Waste to landfill / recycled / composted	3	0.497 / 0.021 / 0.009 kg/kg	DEFRA (indicative)
Commute: car / two-wheeler / bus / metro	3	0.171 / 0.083 / 0.105 / 0.035 kg/km	DEFRA, indicative

5.2 Anomaly Detection

For every electricity meter, the normal profile is learnt separately for each hour of the day and for working and non-working days (weekends, holidays and vacations from the academic calendar) over the previous 28 days. For a recent hourly value x with historical values H in the same bucket, a robust z-score is computed:

$$z = (x - \text{median}(H)) / \max(1.4826 \cdot \text{MAD}(H),\ 0.05 \cdot \text{median}(H),\ 1)$$

An hour is flagged when $z > 4$ and the excess exceeds both 20% of the median and 25 kWh, which suppresses alarms on tiny loads. Flagged hours are grouped by building and operational date (hours after midnight belong to the previous evening). Groups between 22:00 and 06:00 in non-residential buildings become *after-hours* alerts; others become *spike* alerts. Severity is graded on the excess energy, and the estimated cost is excess × tariff.

Leak detection uses the minimum hourly flow between 01:00 and 05:00, when legitimate use is close to zero. If the minimum on a recent night exceeds max(2.5 × baseline, baseline + 0.5 kL/h), where the baseline is the median of the previous four weeks' night minima, one continuing leak work order is opened for that meter.

Data-quality checks raise alerts for six identical non-zero readings (stuck sensor), meters silent for more than three hours (offline) and readings above the meter's physical limit (impossible value). **Indoor environment** alerts are raised when CO_2 exceeds 1200 ppm. Every alert carries a unique de-duplication key, so repeated scans never create duplicates.

5.3 Demand Forecasting

Daily grid electricity y_t is modelled by multiple linear regression fitted with NumPy least squares on the last 120 days:

$$y_t = \beta_0 + \beta_1 \cdot t + \Sigma_{k=1..6}\, \gamma_k \cdot DOW_k + \delta \cdot Holiday_t + \theta \cdot (T_t - 28) + \varepsilon_t$$

where DOW are day-of-week indicators (Sunday as reference), Holiday marks weekday holidays and vacations, and T is the mean outdoor temperature. The coefficient θ directly gives the cooling sensitivity in kWh per degree. Future days use the calendar and the recent average temperature; the 95% prediction interval is ±1.96σ of the residuals. Accuracy is reported as MAPE and R^2.

5.4 Benchmarking and Scenario Modelling

Buildings are compared on the Energy Performance Index, EPI = annualised kWh / floor area (m^2), together with kWh and litres of water per occupant per day. Star bands are modelled on BEE building rating ranges.

The scenario model starts from annualised grid consumption E (from the last 30 days) and applies transparent engineering assumptions that the user can change: LED retrofit saves 50% of the lighting share (20% of E) for the retrofitted fraction; rooftop solar yields 1400 kWh per kWp per year; each 1 °C increase in AC set-point saves 6% of the cooling share (45% of E); occupancy controls save 15% of cooling and lighting in covered areas; an electric bus replaces diesel at 4 km/L but adds 1.1 kWh/km of charging. For each measure the model reports kWh and diesel saved, tCO_2e avoided, annual saving, capital cost and simple payback = capex / annual saving.

CHAPTER 6
IMPLEMENTATION

6.1 Project Structure

The backend is a Django project with four applications. **accounts** holds the custom user model with roles, registration and permission classes. **monitoring** models buildings, gateways, meters, readings, indoor and weather data, the academic calendar and alerts, and implements the ingestion endpoint. **sustainability** stores emission factors, fuel and refrigerant logs, waste weighings, commute surveys, trees and targets. **analytics** contains the services described in Chapter 5, the REST views, and two management commands: `seed_demo`, which builds a realistic campus with 120 days of hourly history, and `simulate`, which acts as a live gateway. The frontend is a React and TypeScript single-page application organised into pages, shared components, an API client and small state stores.

6.2 Carbon Engine

The `FactorBook` class loads every factor version once and answers which value applied on a given day; the engine then produces one line item per building, day and activity. All dashboards, the PDF report and the CSV export are computed from the same line items, which guarantees that the numbers agree everywhere.

```
class FactorBook:
    """Loads every factor version once; answers 'which value applied on this day?'."""
    def __init__(self):
        self._rows = defaultdict(list)
        for factor in EmissionFactor.objects.order_by("valid_from"):
            self._rows[factor.activity].append(factor)

    def get(self, activity, on=None):
        rows = self._rows.get(activity)
        if not rows:
            return None
        on = on or timezone.localdate()
        chosen = rows[0]
        for factor in rows:
            if factor.valid_from <= on:
                chosen = factor
            else:
                break
        return chosen
```

6.3 IoT Ingestion and Simulation

Gateways post batches of up to 5000 readings to `/api/ingest/`. Each reading carries the meter serial, the consumption during the interval and optionally the interval length and timestamp. The endpoint authenticates the gateway key, rejects unknown meters, non-numeric or future values with a per-item error, flags values above the meter's hourly limit as suspect, stores the rest in one bulk insert and updates each meter's last-seen time.

```
POST /api/ingest/        X-Gateway-Key: 6e73...bc69
{"readings": [{"serial": "EM-ACA", "value": 42.5, "interval_minutes": 15},
              {"serial": "WM-BH1", "value": 0.61, "interval_minutes": 15}]}
-> 201 {"accepted": 2, "rejected": []}
```

Because physical meters were not available, the simulator models each building from first principles: an always-on base load, an occupancy-driven load that follows the timetable of its building type, and a cooling load proportional to the outdoor temperature above 24 °C. Solar output follows irradiance with day-to-day cloud cover, water use follows daily routines, and the weather is Chennai-like with monsoon rain in October. Day-level randomness is seeded by date, so historical and live data agree. The seed command deliberately injects six faults (air-conditioning left on for two nights, a hostel tank leak, a lab test rig left on, a poorly ventilated hall and an impossible meter value) to validate detection.

6.4 User Interface

The web application provides thirteen screens grouped into Overview, Modules, Intelligence and Administration. Server state is cached with TanStack Query; the live console refreshes every 30 seconds. Charts use Recharts and the campus map uses Leaflet with free OpenStreetMap tiles. The layout adapts to mobile screens, and pages are code-split so that the initial download stays small.

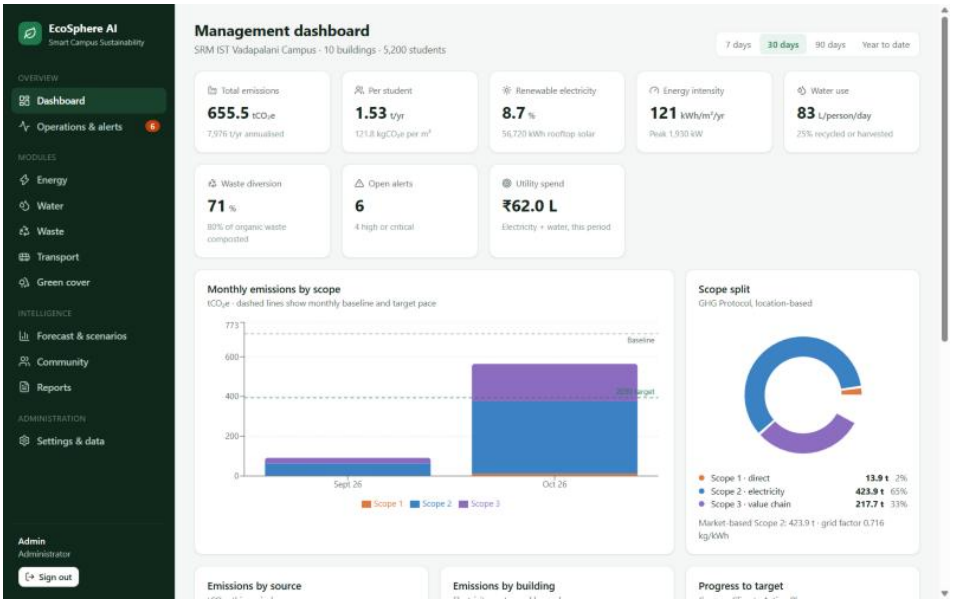


Figure 6.1: Management dashboard with KPIs, monthly scope trend and target reference lines

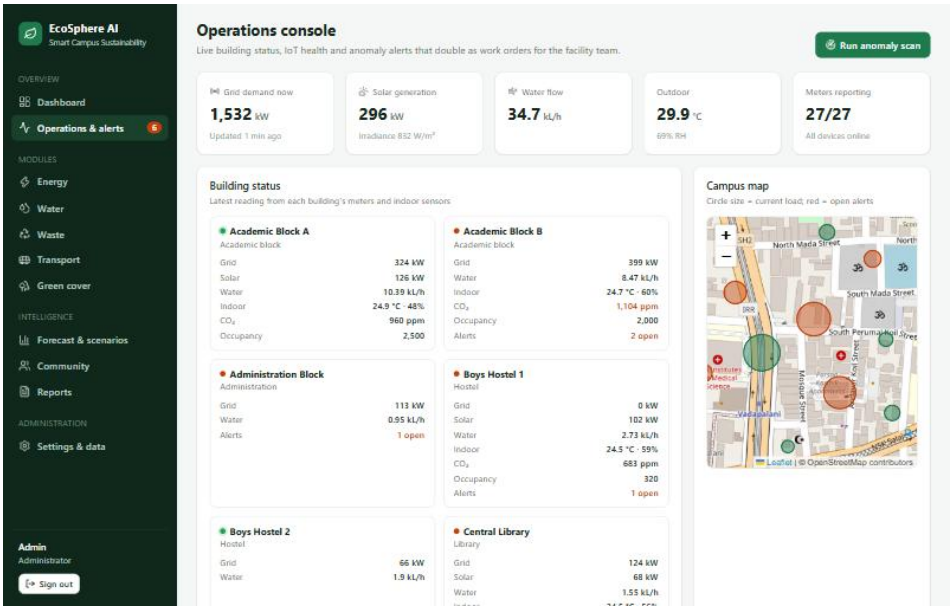


Figure 6.2: Operations console: live building status, IoT health and campus map

CHAPTER 7
RESULTS AND DISCUSSION

7.1 Campus Footprint

Table 7.1 summarises the key performance indicators reported by the system for the most recent 30 days of the evaluation dataset (ten buildings, 65,500 m^2, 5,200 students and 650 staff).

Table 7.1: Campus KPIs reported by the system (last 30 days)

Indicator	Value	Indicator	Value
Total emissions (30 days)	655.5 tCO_2e	Renewable share of electricity	8.7%
Annualised emissions	7,975 tCO_2e/yr	Energy Performance Index	120.5 kWh/m^2/yr
Per student	1.53 t/yr	Peak demand	1,930 kW
Per floor area	121.8 kg/m^2/yr	Water use	82.5 L/person/day
Scope 1 / 2 / 3	13.9 / 423.9 / 217.7 t	Recycled + harvested water	25.0%
Electricity cost (30 days)	Rs. 53.3 lakh	Waste diversion rate	71.3%
Tree sequestration	20.2 tCO_2/yr	Sustainable commute share	64.0%

Purchased electricity dominates the footprint at about 65% of the total, as expected for a cooling-dominated campus in South India, followed by student and staff commuting. Direct Scope 1 sources are small, but refrigerant leakage stands out: a single 6 kg R-410A top-up equals about 12.5 tCO_2e, more than a month of the entire bus fleet. Tree sequestration offsets only about 0.25% of annual emissions, which confirms that planting is not a substitute for reducing consumption.

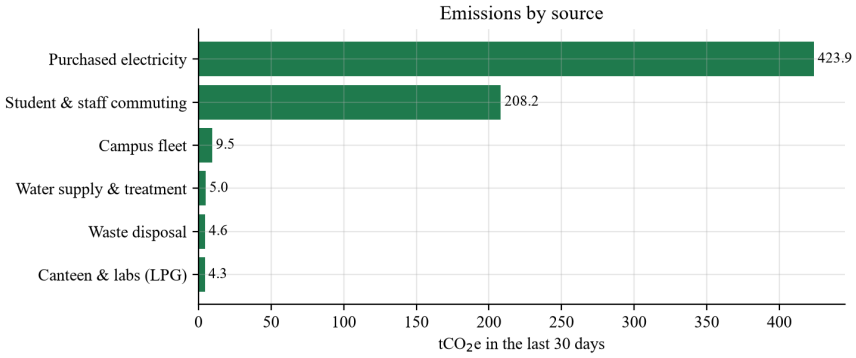


Figure 7.1: Emissions by source over the last 30 days

7.2 Energy Use Patterns

Building-level metering reveals patterns that a single main meter hides. Academic blocks and laboratories follow the timetable, with laboratories retaining a high base load from equipment; hostels peak in the evening; the food court shows breakfast, lunch and dinner peaks. Rooftop solar on Academic Block A visibly shaves the midday grid draw.

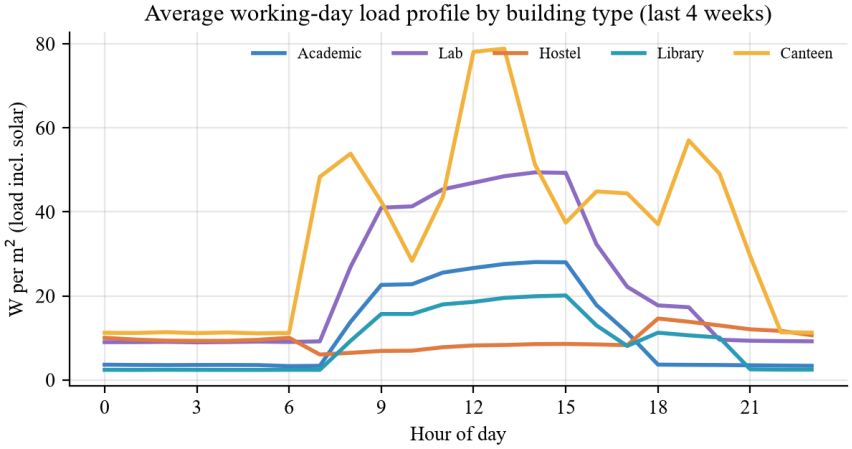


Figure 7.2: Average working-day load profile by building type

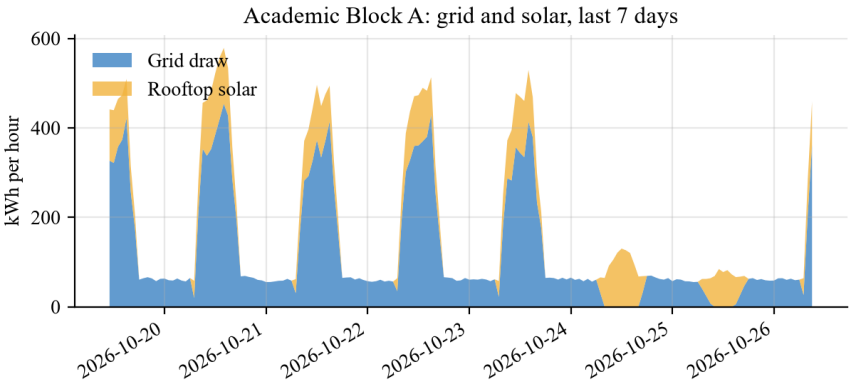


Figure 7.3: Grid draw and rooftop solar generation, Academic Block A

7.3 Fault Detection

The anomaly scan raised exactly six alerts on the evaluation dataset, one for each injected fault, with no false positives on the 120 days of normal data. Figure 7.4 shows the after-hours alert for Academic Block B: night-time load rose from about 50 kWh to over 200 kWh per hour for two nights while the learnt profile stayed flat, an excess of roughly 1,290 kWh (about Rs. 11,600) per night. Figure 7.5 shows the leak signature in Boys Hostel 1, where minimum night flow jumped from about 0.2 to 2.6 kL/h, an estimated loss of 58 kL per day. In a manual, bill-based process such faults would typically surface only weeks later.

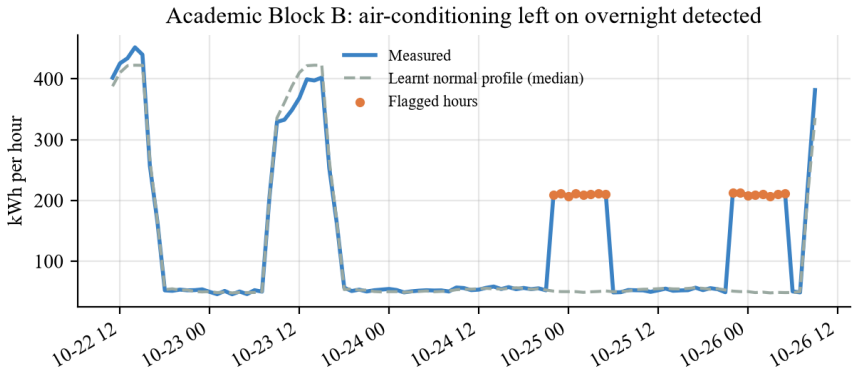


Figure 7.4: After-hours air-conditioning load detected in Academic Block B

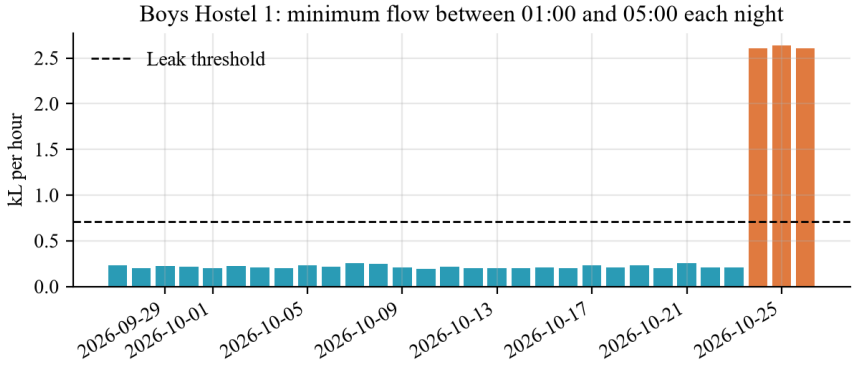


Figure 7.5: Night-time minimum flow revealing a leak in Boys Hostel 1

Table 7.2: Detection results for injected faults

Injected fault	Detected as	Severity	Estimated impact
AC left on overnight (two nights), Academic Block B	After-hours load x2	High	~1,290 kWh per night
Leaking tank float valve, Boys Hostel 1	Possible water leak	High	~58 kL per day
Test rig left at full power, Tech Park Labs	Electricity spike	High	~780 kWh
Packed hall with fresh air off, Central Library	Poor ventilation	Medium	CO_2 ~1,490 ppm
99,999 kWh reading, Administration Block	Impossible value	Low	Flagged, excluded

7.4 Forecasting Accuracy

The regression model was trained on 120 days and achieved a mean absolute percentage error of 2.36% and R^2 of 0.987. It captures the weekly cycle, the lower demand during holidays and vacations, and a cooling sensitivity of about 935 kWh per day for each additional degree Celsius. The 30-day forecast predicts 614 MWh, 439 tCO_2e and a bill of about Rs. 55.2 lakh, which supports budgeting and peak-load planning.

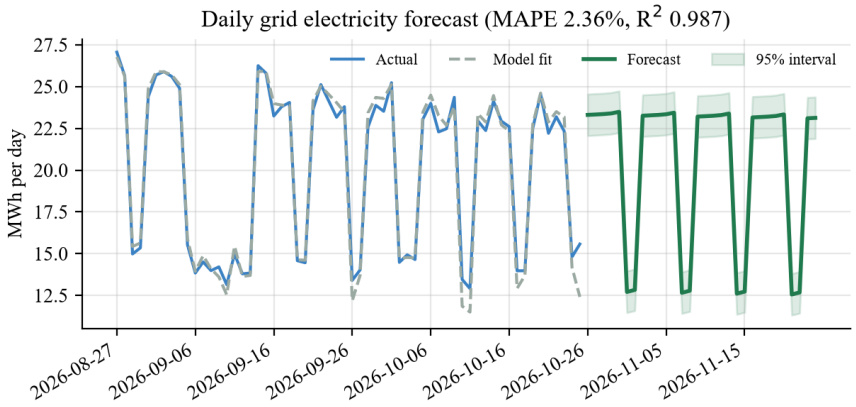


Figure 7.6: Actual, fitted and forecast daily grid electricity

7.5 Benchmarking

The food court has the highest Energy Performance Index (255 kWh/m^2/yr) because of cooking and refrigeration in a small floor area, followed by Tech Park Labs (157). The sports complex is the most efficient. Normalising by area and occupancy makes such comparisons fair and points retrofit budgets at the right buildings.

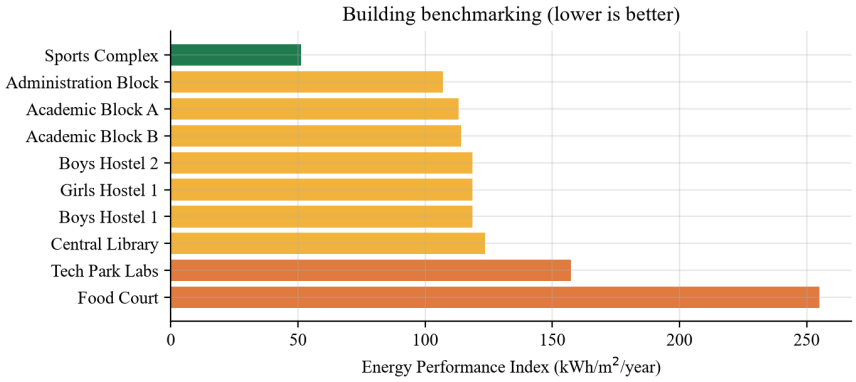


Figure 7.7: Energy Performance Index by building

7.6 Waste, Transport and Engagement

The waste module shows a diversion rate of 70.1%, with 80% of organic waste composted or converted to biogas, and the weekly trend in Figure 7.8 reflects the improvement from a segregation drive. Diverting waste from landfill avoided about 42.02 tCO_2e over the period. Commute surveys show 64% of trips by walking, cycling, bus, metro or electric two-wheeler, yet commuting still contributes about 2,533 tCO_2e per year, mostly from cars and petrol two-wheelers. The hostel leaderboard converts week-on-week reductions into points, giving residents visible feedback.

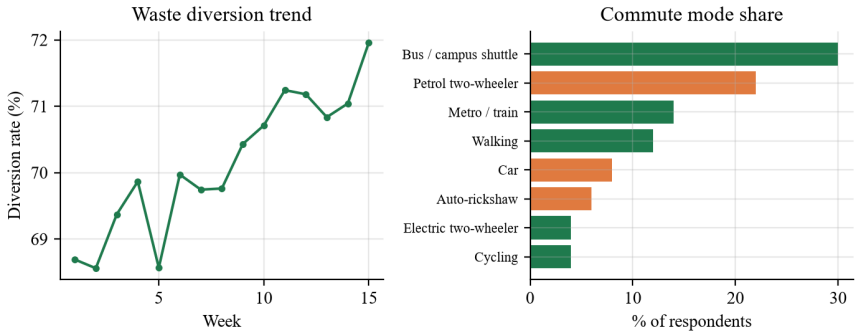


Figure 7.8: Weekly waste diversion rate and commute mode share

7.7 Scenario Analysis

Table 7.3: Scenario: 50% LED retrofit, 250 kWp solar, +1 °C set-point, 30% occupancy controls, 2 e-buses

Measure	tCO_2e/yr	Saving/yr	Capex	Payback
LED lighting retrofit	260	Rs. 32.7 L	Rs. 39.3 L	1.2 yr
Rooftop solar expansion	251	Rs. 31.5 L	Rs. 112.5 L	3.6 yr
Air-conditioning set-point increase	140	Rs. 17.7 L	Nil	Immediate
Occupancy-based HVAC & lighting controls	152	Rs. 19.1 L	Rs. 19.6 L	1.0 yr
Electric campus buses	-6	Rs. 6.8 L	Rs. 160.0 L	23.5 yr
Total	**797**	**Rs. 107.8 L**	**Rs. 331.4 L**	**3.1 yr**

The combined package would cut grid electricity by 14.7% and avoid about 797 tCO_2e per year with a payback of 3.1 years. Raising air-conditioning set-points costs nothing, and LED retrofits and occupancy controls pay back in about a year. The model also exposes a counter-intuitive result: on today's coal-heavy grid, electric buses save money but slightly *increase* emissions unless they are charged from solar, an insight that would be easy to miss without quantitative modelling.

CHAPTER 8
TESTING AND VALIDATION

8.1 Testing Strategy

Testing combined automated end-to-end tests, manual interface testing for every role and screen, and data validation against known inputs. The automated suite seeds a fresh 45-day campus in an isolated test database and exercises the carbon engine, anomaly detection, ingestion, permissions and every analytics endpoint. The frontend is type-checked by the TypeScript compiler on every build.

8.2 Test Cases

Table 8.1: Automated test cases (18 of 18 passing)

#	Test case	Expected result	Status
1	Electricity line item equals kWh × 0.716	Exact match for every building-day	Pass
2	New factor version from a given date	Old value before, new value after	Pass
3	Suspect 99,999 kWh reading	Excluded from totals	Pass

#	Test case	Expected result	Status
4	All three scopes present in 30-day inventory	Scopes 1, 2 and 3 reported	Pass
5	Injected faults	After-hours, spike, leak, CO_2 and impossible-value alerts	Pass
6	Repeated anomaly scan	No duplicate alerts	Pass
7	Ingest batch with bad items	2 accepted, 2 rejected with reasons, 1 flagged	Pass
8	Ingest with wrong gateway key	HTTP 401	Pass
9	Login and profile	Token issued; role returned	Pass
10	Anonymous access to analytics	HTTP 401	Pass
11	Student writes factors, logs, alerts, users	HTTP 403 for each	Pass
12	Facility resolves an alert	Status resolved, timestamp set	Pass
13	Administrator demotes self	HTTP 400, refused	Pass
14	Commute answers privacy	User sees only own answers	Pass
15	All 16 analytics/report endpoints	HTTP 200	Pass
16	Invalid date ranges and datasets	HTTP 400 with message	Pass
17	14-day forecast	14 points, MAPE below 10%	Pass
18	GHG report download	Valid PDF document	Pass

8.3 Performance and Usability

With about 78,000 hourly readings, the campus overview endpoint responds in about 0.2 s, a year-to-date overview in under 1 s and the GHG PDF in about 1 s on an ordinary laptop. Seeding 120 days of data takes about seven seconds. Manual tests confirmed that each role sees only the navigation and actions it is allowed, that the live console updates as the simulator streams readings, that forms validate input and show clear error messages, and that the layout works without horizontal scrolling on a 375-pixel mobile screen.

CHAPTER 9
CHALLENGES AND FUTURE ENHANCEMENTS

9.1 Challenges and Mitigation

Table 9.1: Challenges and mitigation

Challenge	Mitigation adopted
High upfront cost of metering	Free software stack; phased, building-by-building rollout; ESP32-class low-cost sensing where accuracy needs are modest
Legacy and incompatible meters	Gateway adapter pattern; simple HTTP ingestion that any device can call
Poor data quality	Validation, suspect flags, stuck/offline checks, completeness KPIs, cross-checks with utility bills
Changing emission factors	Dated, versioned factor register; append-only history
Privacy of personal travel data	Anonymous aggregation; individuals see only their own answers
Low community engagement	Leaderboards, points, personal footprint and practical tips

9.2 Limitations

- The evaluation uses simulated data; accuracy must be re-validated against real meters and bills during a pilot.
- Default emission factors are indicative and must be replaced with the latest official values.
- SQLite suits a pilot; a campus-wide deployment with minute-level data should move to PostgreSQL with TimescaleDB, which the ORM supports without code changes.
- Scope 3 excludes purchased goods and business travel.

9.3 Future Enhancements

- AI-assisted control that adjusts air-conditioning set-points and lighting automatically from occupancy and weather.
- A digital twin of the campus for testing interventions virtually.
- Native MQTT and LoRaWAN adapters and WebSocket push for sub-minute live views.
- Integration with an internal carbon price or green fund that reinvests verified savings.
- Tamper-evident (hash-chained) logs for auditable carbon records.
- Smart-grid features such as demand response and battery storage coordination.
- A shared regional platform so that neighbouring institutions can benchmark against each other.

CHAPTER 10
CONCLUSION

This project delivered a complete, working Smart Campus Sustainability and Carbon Footprint Monitoring System that gives an institution what annual reports cannot: timely, granular and trustworthy information about its environmental impact. IoT readings flow through an authenticated ingestion layer into a standards-based carbon engine; robust anomaly detection finds wastage within hours; forecasting, benchmarking and scenario modelling turn data into decisions; and role-specific dashboards involve the whole campus community.

On the evaluation campus the system quantified an annualised footprint of about 7,975 tCO_2e, detected every injected fault without false alarms, forecast demand within 2.36% error and identified a package of measures worth about 797 tCO_2e per year. Because it is built entirely on free and open-source software and runs on an ordinary computer, it can begin as a low-risk pilot in two or three buildings and grow building by building, exactly as the phased roadmap of the project report recommends.

Beyond its own savings, a monitored campus demonstrates to students and to society how data-driven climate action can work in practice, and the live data strengthens the institution's educational mission.

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APPENDIX: INSTALLATION AND EXECUTION

Requires Python 3.11+ and Node.js 20+ (both free). On Windows, run `setup.bat` once, then `start.bat`, which opens the app at http://localhost:5173; commands for other systems are in README.md. Demo logins: **admin** / admin123, **facility** / facility123, **student** / student123.